

Learning classification on hierarchical ultrametric domains: generalization bounds in the p -adic setting

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Abstract

Many data domains are naturally hierarchical, with similarity governed by taxonomy-like or tree-structured relations rather than by Euclidean geometry. Ultrametric spaces provide a principled mathematical model for such settings, but their role in statistical learning theory remains comparatively underdeveloped. In this paper, we study binary classification on the ultrametric domain of p -adic integers, viewed as a mathematically tractable model of hierarchical structure. We consider concept classes given by finite unions of p -adic balls and introduce a signed-width representation that places these concepts within a margin-based learning framework. This formulation yields a tractable analysis of capacity and generalization: we show that the underlying class of distance functions has pseudo-dimension 1, derive covering-number bounds, and obtain non-asymptotic generalization guarantees for large-width classification. The results highlight how the geometry of ultrametric domains can lead to remarkably simple complexity behavior, despite the richness of the underlying concept classes. Beyond the specific p -adic setting, the paper provides a mathematically explicit framework for studying classification and generalization in hierarchical ultrametric settings.

Keywords: learning classification, large margin, ultrametric spaces, p -adic integers

1 Introduction

Many learning problems involve data whose similarity structure is inherently hierarchical rather than Euclidean. Examples include phylogenetic data, taxonomies, ontologies, directory trees, and other domains in which proximity is governed by ancestry, refinement, or common descent. In such settings, the relevant geometry is often better captured by an ultrametric than by a Euclidean metric. This motivates the study of classification and generalization in ultrametric domains.

Ultrametric spaces are metric spaces in which the distance satisfies the strong triangle inequality: for any three points x, y, z ,

$$d(x, y) \leq \max \{d(x, z), d(z, y)\}. \quad (1)$$

Their geometry is naturally hierarchical: balls are nested, and triangles are isosceles with the two longer sides equal. As a result, ultrametric spaces provide a principled mathematical model for hierarchical similarity.

Examples arise in phylogenetic trees [1, 2], p -adic analysis [3], and computer vision [4]. A rooted tree or dendrogram induces an ultrametric, and conversely the family of balls in an ultrametric space reflects

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<https://github.com/ratsaby/joel-ratsaby-preprints/tree/main/p-adic-concepts>

a hierarchy. Hierarchical data often appears in the form of a taxonomy; one example is WordNet [5], whose synonym sets are arranged through semantic relations.

Distances induced by taxonomies are not always represented well by Euclidean embeddings. This has led to growing interest in methods that use hierarchical distance information directly. Recent work includes neural architectures in ultrametric p -adic space [6], low-distortion ultrametric embeddings of Euclidean data [7], learning ultrametric trees for optimal transport regression [8], incremental treatment of tree metrics and ultrametrics in data streams [9], and ultrametric feature spaces for hierarchical segmentation [4]. Despite this growing body of work, the learning-theoretic analysis of classification in ultrametric domains remains relatively limited.

In statistical learning theory, one basic principle is that classifiers that separate oppositely labeled data with a large margin tend to generalize better [10–14]. A generalization of the large-margin idea to metric spaces and, more generally, to distance spaces is the framework of learning large *width* classifiers, introduced in [15], extended in [16–25] to various kinds of input domains and classes of functions, and implemented as a learning algorithm [26]. The large-width framework provides a natural way to quantify separation in such spaces and is therefore well suited to hierarchical settings.

Consider a sample of labeled points $\{(x_i, y_i)\}_{i=1}^m$, with $x_i \in X$, classification labels $y_i \in Y := \{-1, 1\}$, $1 \leq i \leq m$, and an input space X with distance function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ where $\mathbb{R}_{\geq 0}$ denotes the set of non-negative real numbers. A *classifier* is a function $h : X \rightarrow Y$. Let S_h^+ be a subset of X classified as 1 and S_h^- a subset classified as -1 , by h . If x is a sample point with a label $y = 1$ then we define the classification width at x to be the distance $d(x, S_h^-)$ defined as the minimum (or infimum, in the case that S_h^- is infinite) of the distances between x and elements of S_h^- . Similarly, if x is labeled -1 then the width at x is defined as the distance $d(x, S_h^+)$. The sample width is defined to be the minimum, over the sample points, of these individual widths.

The generalization error bounds obtained in [15–25] improve as the sample width increases. This motivates the study of large-width classification as a framework for understanding generalization in structured metric domains, including ultrametric ones.

In this paper we study binary classification on the ultrametric domain $\mathbb{Z}_p$ of p -adic integers, where p is a prime number. The space $\mathbb{Z}_p$ is infinite, compact, and admits a natural rooted-tree representation in which each node has a unique parent, except the root, while p -adic integers may be viewed as leaves at infinite depth, that is, boundary points of the infinite tree [27]. It therefore provides a canonical and mathematically tractable model of hierarchical structure.

The space $\mathbb{Z}_p$ is of course a standard object in p -adic analysis and number theory [3, 27, 28]. For the purposes of this paper, however, its importance lies in the fact that it is equipped with the p -adic metric, with respect to which it is an ultrametric space, and that its balls admit an explicit description as residue classes modulo powers of p . An element of $\mathbb{Z}_p$ may be viewed as an infinite sequence of digits in $\{0, \dots, p-1\}$, and a ball is determined by a finite prefix of this sequence. This makes $\mathbb{Z}_p$ a convenient concrete model for hierarchical domains, in which geometric notions such as distance and width can be expressed through explicit symbolic and arithmetic descriptions of concepts.

We consider classifiers based on concepts that are finite unions of maximal balls in $\mathbb{Z}_p$. To analyze them, we use a signed-width representation that places these concepts within a margin-based real-valued framework. This leads to a tractable analysis of capacity and generalization for classification on a hierarchical ultrametric domain.

1.1 Contribution

The contributions of the paper are as follows. First, we formalize a concept class on $\mathbb{Z}_p$ consisting of finite unions of maximal p -adic balls. Since balls in $\mathbb{Z}_p$ correspond to residue classes modulo powers of p , this yields an explicit and mathematically tractable model for hierarchical classification in an ultrametric domain.

Second, we introduce a signed-width representation for these concepts. The representation associates with each classifier a real-valued signed-width function whose sign recovers the classifier itself. This embeds the classification problem into a margin-based learning framework and makes it possible to analyze generalization through notions of width and margin, extending the large-width approach of [15, 25] to the p -adic setting. In this representation, the width at a point is determined by the hierarchical position of the corresponding ball in the tree: larger width corresponds to coarser balls and therefore to simpler residue-class descriptions of the concept.

Third, we derive upper bounds on the complexity of the induced function classes. A key observation is that the ultrametric structure of $\mathbb{Z}_p$ imposes a strong combinatorial regularity on balls: any two balls are either nested or disjoint. Using this property, we show that the combinatorial dimension of $\mathbb{Z}_p$ equals 1, which in turn implies that the associated class of distance functions has pseudo-dimension 1. This yields especially simple covering-number bounds compared to those arising in more general metric settings.

Fourth, we develop a covering-number analysis for the class of signed-width functions associated with unions of at most n balls. The proof combines the low pseudo-dimension of the underlying distance class with a combinatorial counting argument based on dendrogram partitions of finite samples. More specifically, intersections of p -adic balls with a finite sample induce a hierarchical clustering structure, and the number of admissible partitions generated by these blocks controls the size of the approximating family. This leads to explicit covering-number bounds for the signed-width class.

Finally, we apply these complexity bounds to obtain non-asymptotic generalization guarantees for large-width classification on $\mathbb{Z}_p$. The resulting bounds hold uniformly over the width parameter and show that, for classifiers consistent with the sample, the error bound improves as the sample width increases. Thus, beyond the specific p -adic setting, the paper provides a mathematically explicit framework for studying classification and generalization on hierarchical ultrametric domains.

The paper is organized as follows: Section 2 reviews the required background on ultrametric spaces and the p -adic integers $\mathbb{Z}_p$, including their tree representation, ultrametric geometry, and the structure of p -adic balls. Section 3 introduces the class of p -adic concepts defined as finite unions of maximal balls and the associated binary classifiers. Section 4 develops the notion of classification width and signed width in the p -adic setting, establishes their connection to the hierarchical structure of the underlying concepts, and studies maximal-width classifiers on finite samples. Section 5 analyzes the combinatorial complexity of the induced function classes, deriving bounds on combinatorial dimension, pseudo-dimension, and covering numbers. Section 6 constructs approximating families for the signed-width function class and obtains covering-number bounds for these classes. Finally, Section 7 applies these results to statistical learning theory, deriving non-asymptotic large-width generalization bounds for classification on $\mathbb{Z}_p$, while Section 8 concludes with a discussion of the implications of the results and directions for future research.

2 Background

2.1 Ultrametric space

Let (X, d) be a metric space.

Definition 1. (*Ultrametric Space*) A metric space (X, d) is called an *ultrametric space* if for any $x, y, z \in X$, $d(x, y) \leq \max \{d(x, z), d(z, y)\}$. This inequality is called the *strong triangle inequality*.

Proposition 2. (*Isosceles property*) Let (X, d) be an ultrametric space. For any $x, y, z \in X$, if $d(x, z) < d(y, z)$, then $d(x, y) = d(y, z)$.

Proof. By the strong triangle inequality,

$$d(x, y) \leq \max \{d(x, z), d(z, y)\} = d(y, z). \quad (2)$$

On the other hand, $d(y, z) \leq \max \{d(y, x), d(x, z)\}$. Since $d(x, z) < d(y, z)$, the maximum on the right cannot be $d(x, z)$. Therefore it must be $d(y, x)$, and hence $d(y, z) \leq d(y, x) = d(x, y)$. Combining the latter with (2) gives $d(x, y) = d(y, z)$. $\square$

This is called the Isosceles property because in a triangle with vertices x, y, z , the two longer sides satisfy $d(x, y) = d(y, z)$.

A ball in X is defined as the set of all points whose distance from a center point is no more than some constant, which is referred to as its radius. An open ball of radius r centered at $a \in X$ consists of all points $z \in X$ whose distance from a is less than r . A closed ball of radius r centered at $a \in X$ consists of all points $z \in X$ whose distance from a is less than or equal to r .

Proposition 3. [3, Proposition 18.5] *Let B_1 and B_2 be balls in an ultrametric space X . Then either B_1 and B_2 are ordered by inclusion, $B_1 \subseteq B_2$ or $B_2 \subseteq B_1$, or B_1 and B_2 are disjoint.*

Proposition 4. [3] *Let B be a ball in an ultrametric space X . Then B is both open and closed in the topology induced by the metric.*

2.2 $\mathbb{Z}_p$

For an integer $n \geq 2$, let

$$\mathcal{I}_n := \{0, 1, \dots, n-1\}.$$

Following [3, Definition 3.1], who defines the space $\mathbb{Z}_n$ of n -adic integers, let p denote a prime number and define $\mathbb{Z}_p$ to be the set of all infinite sequences

$$(a_0, a_1, \dots) \tag{3}$$

where each a_i belongs to $\mathcal{I}_p$. The elements of $\mathbb{Z}_p$ are p -adic integers, and $\mathbb{Z}_p$ forms a ring under the standard p -adic addition and multiplication operations.

If for a finite integer $N \geq 0$ the coefficients in the expansion of x , satisfy $a_i = 0$, for all $i \geq N$, then x is a non-negative integer (a natural number). Therefore the set $\mathbb{N}$ of natural numbers is a subset of the set of p -adic integers $\mathbb{N} \subset \mathbb{Z}_p$. A negative integer has a tail $a_i = p-1$, $i \geq M$, starting from some finite M .

For a p -adic integer $x \in \mathbb{Z}_p$ let $\nu_p(x)$ denote its *valuation*, or *order* [3, Definition 3.4], which is defined as a non-negative integer k such that $x = p^k c$ where $p \nmid c$. The valuation of a p -adic integer x is zero if p does not divide x . Equivalently, the valuation can be expressed as

$$\nu_p(x) = \begin{cases} \max \{k : p^k \mid x\} & \text{if } x \neq 0 \\ \infty & \text{if } x = 0. \end{cases}$$

The valuation measures the unique exponent of p dividing x .

The p -adic absolute value of x is defined by

$$|x|_p := \frac{1}{p^{\nu_p(x)}}$$

where $|0|_p = 1/p^\infty = 0$. It follows that for all $x \in \mathbb{Z}_p$,

$$|x|_p \leq 1 \tag{4}$$

because $\nu_p(x) \geq 0$.

Definition 5. [29, p. 9](Distance) For $x, z \in \mathbb{Z}_p$, define the p -adic distance by

$$d(x, z) := |x - z|_p. \quad (5)$$

The distance (5) satisfies the strong triangle inequality (1) hence it is an ultrametric. Therefore we have the following proposition.

Proposition 6. The space $\mathbb{Z}_p$ equipped with the p -adic distance (5) is an ultrametric space.

Corollary 7. The strong triangle inequality (1) and the Isosceles property (Proposition 2) hold for $\mathbb{Z}_p$.

The tree $\mathbb{T}_p$ corresponding to $\mathbb{Z}_p$ is defined as follows.

Definition 8. (Tree) An infinite tree $\mathbb{T}_p$ corresponds to $\mathbb{Z}_p$. It has a single root at level -1 , and nodes at infinitely many levels $k \in \{0, 1, \dots\}$. At the k^{th} level there are p^{k+1} nodes $a = [a_0, a_1, \dots, a_k]$, where $a_i \in \mathbb{I}_p$, $0 \leq i \leq k$, and the nodes are divided into p^k disjoint groups of nodes. Each group consists of p sibling nodes. For $k \geq 1$, the sibling nodes in each group share the k common ancestor nodes (excluding the root) $[a_0], [a_0, a_1], \dots, [a_0, a_1, \dots, a_{k-1}]$, where the deepest common ancestor $[a_0, a_1, \dots, a_{k-1}]$ is their parent and is denoted $\pi(a) = [a_0, a_1, \dots, a_{k-1}]$. For $k = 0$, the p nodes $[a_0], a_0 \in \mathbb{I}_p$, have the root as their common parent, and for each such node a we define $\pi(a)$ to be the root. The p -adic integers $x = (x_0, x_1, \dots)$ correspond to the leaves of $\mathbb{T}_p$. A node (or leaf) is called *descendant* of a node a if a is one of its ancestors. Every node except the root has a unique parent and every node is connected to the root by a unique finite path. Hence $\mathbb{T}_p$ is a rooted tree.

Definition 9. (Level) We denote by $\ell(a) \geq 0$ the level of a node a in $\mathbb{T}_p$. The level of the root is defined to be -1 .

Figure 1 displays an infinite tree $\mathbb{T}_3$ representation of $\mathbb{Z}_3$ where 3-adic integers are represented by leaves (not seen since they are infinitely deep). Given two 3-adic integers $x = (x_0, x_1, \dots)$ and $z = (z_0, z_1, \dots)$, suppose for example that $x_0 = z_0 = a_0$ and $x_1 \neq z_1$, then the deepest common ancestor is their parent node $[a_0]$ at level 0. Hence the valuation of their difference $\nu_3(x - z) = 1$ and therefore the distance $|x - z|_3 = 1/3$. Two unique nodes at level 1 correspond to two disjoint balls B and B' . Any p -adic integers $x, z \in B$ and $x' \in B'$ satisfy $\nu_3(x - z) = 2$ and $\nu_3(x - x') \leq 1$.

In general, at level $k \in \{0, 1, \dots\}$, a node has p children in level $k + 1$. Each such child corresponds to a choice of the next digit $a_{k+1} \in \{0, \dots, p - 1\}$ in the p -adic expansion of an integer. At level k , every node of $\mathbb{T}_p$ is a vector

$$[a_0, a_1, \dots, a_k] \in \mathbb{I}_p^{k+1} \quad (6)$$

which represents a *residue class* (or a coset)

$$\sum_{i=0}^k a_i p^i + p^{k+1} \mathbb{Z}_p = \left\{ x \in \mathbb{Z}_p : x \equiv \sum_{i=0}^k a_i p^i \pmod{p^{k+1}} \right\}. \quad (7)$$

This residue class is a closed *ball* of radius $1/p^{k+1}$ that consists of all p -adic integers whose p -adic expansion starts with $a_0, a_1, \dots, a_k$.

Note that $[a_0, a_1, \dots, a_k]$ is *not* a p -adic integer since it is not an infinite sequence of the form (3). If we append a trail of zeros to it then it is a p -adic integer (and also, a natural integer).

That Proposition 3 holds for $\mathbb{Z}_p$ follows from the fact that two balls satisfy $B \subseteq B'$ if and only if the corresponding node a' of B' is an ancestor of the corresponding node a of B in the tree $\mathbb{T}_p$, otherwise they are disjoint.

Definition 10. (LCA) For x and z in $\mathbb{Z}_p$, the *Least Common Ancestor* node $LCA(x, z)$ is their deepest (maximum level) common ancestor.

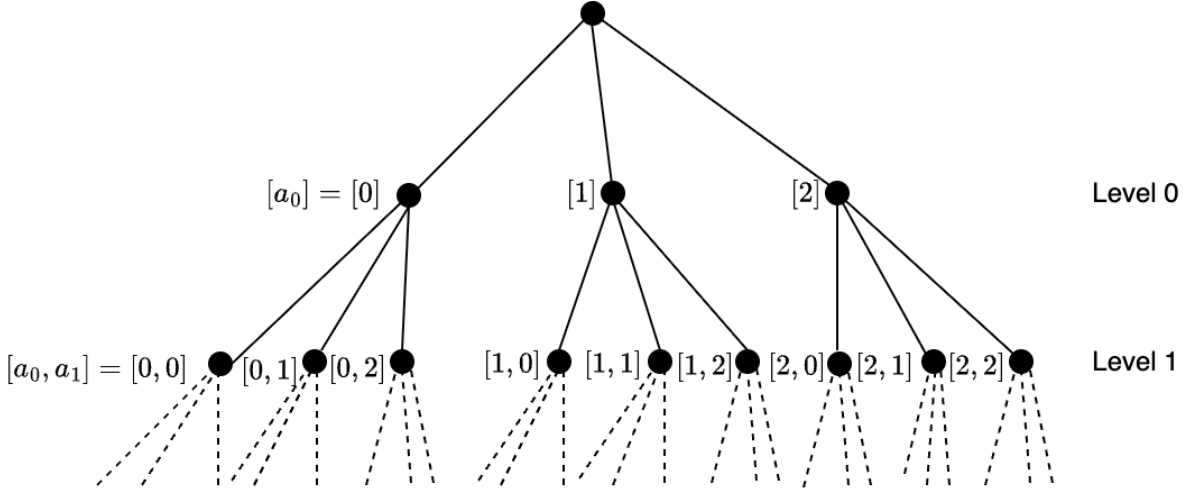


Figure 1: Tree $\mathbb{T}_p$ corresponding to $\mathbb{Z}_p$ ($p = 3$), has infinite number of levels $k \in \{0, 1, \dots\}$. The dashed lines connect to the next level (not shown). Leaves correspond to the p -adic integers and are not shown since their level $k = \infty$.

The word *least* is with respect to the partial order of descendency, where descendant $\leq$ ancestor.

Proposition 11. *The distance (5) between two p -adic integers x and z satisfies*

$$d(x, z) = 1/p^{\ell(LCA(x, z)) + 1}. \quad (8)$$

Definition 12. (*Diameter*) For a subset A of a metric space (X, d) , its diameter is defined by $\text{diam}(A) := \sup\{d(x, y) : x, y \in A\}$.

For any $x, z \in \mathbb{Z}_p$,

$$\ell(LCA(x, z)) \geq -1$$

because by Definition 9, the root has level -1 . Therefore from (8), we have

$$\begin{aligned} 1/p^{\ell(LCA(x, z)) + 1} &\leq 1/p^{-1 + 1} \\ &= 1. \end{aligned}$$

Consider for example $x = (0, 0, \dots)$ and $z = (1, 1, \dots)$. At every level $l > -1$, they do not share a common ancestor, hence $\ell(LCA(x, z)) = -1$. Hence there exists a pair whose distance is 1, and therefore

$$\text{diam}(\mathbb{Z}_p) = 1. \quad (9)$$

Definition 13. (*LCA*) For a node $a \in \mathbb{T}_p$ and $x \in \mathbb{Z}_p$, the Least Common Ancestor node is denoted by $LCA(x, a)$. If a is an ancestor of x then $LCA(x, a) = a$.

Next, we define a tree-based distance quantity between a node and a p -adic integer. For notational simplicity, we continue to use the symbol d for this quantity.

Definition 14. For a node $a \in \mathbb{T}_p$ and $x \in \mathbb{Z}_p$, the distance $d(x, a)$ between x and a is defined as

$$d(x, a) := 1/p^{\ell(LCA(x, a)) + 1}. \quad (10)$$

It follows from the above definitions that if a is an ancestor of x then $d(x, a) = 1/p^{\ell(a) + 1}$.

Denote by

$$\mathbb{Z}_{\geq 0} := \mathbb{N} \cup \{0\}$$

the set of non-negative integers. Define

$$J_p := \{1/p^k : k \in \mathbb{Z}_{\geq 0}\}$$

and let $d_a : \mathbb{Z}_p \rightarrow J_p$ be

$$d_a(x) := d(x, a). \quad (11)$$

2.3 Ball

For a p -adic integer x and a rational fraction $r = 1/p^k$, where $k \in \mathbb{Z}_{\geq 0}$, define a p -adic closed *ball* of radius r centered at x to be the set

$$B_r(x) := \{z : |z - x|_p \leq r\}. \quad (12)$$

Remark 15. By [3, Proposition 18.4], every point inside a ball may serve as its center, and every ball in $\mathbb{Z}_p$ is both closed and open.

We use an alternate description of a ball which is based on a node of $\mathbb{T}_p$ instead of a specific center point. For a node

$$a := [a_0, a_1, \dots, a_{k-1}] \in \mathcal{I}_p^k, \quad (13)$$

in $\mathbb{T}_p$, with level $\ell(a) = k - 1$, denote by

$$r_a = 1/p^{\ell(a)+1}, \quad (14)$$

and let z be any p -adic integer whose expansion coefficients satisfy $z_i = a_i$, $0 \leq i \leq k - 1$. Denote by

$$B(a) := \{x : |x - z|_p \leq r_a\}. \quad (15)$$

This yields an alternative description of a ball because for any p -adic integer x and rational $r = 1/p^k$, a ball $B_r(x)$ can be represented as $B(a)$ by letting a be an ancestor node of x with $\ell(a) = k - 1$ and $r = r_a$. Henceforth, in (15), we choose the center z to be $(a_0, \dots, a_{k-1}, 0, \dots)$ which is represented by a leaf descendant of a (we choose this leaf for simplicity, however any of the infinitely many leaves that are descendants of a can be chosen).

For the root node, which has level -1 , we define

$$r_{\text{root}} := 1, \quad B(\text{root}) := \mathbb{Z}_p.$$

Remark 16. For a node $a \in \mathbb{T}_p$, every $x \in B(a)$ is a descendant of a .

We denote by

$$\mathcal{B}_p := \{B(a) : a \in \mathbb{T}_p\} \quad (16)$$

the class of such balls.

For instance, if $a = [1, 2]$ then ball $B(a)$ is centered at p -adic $(1, 2, 0, \dots)$, its radius is $1/p^2$ and it contains any point $z \in \mathbb{Z}_p$ whose expansion satisfies $z_0 = 1$, $z_1 = 2$. If $a' = [1]$ then $B(a')$ is centered at $(1, 0, \dots)$, its radius $r = 1/p$ and it contains any point $z \in \mathbb{Z}_p$ whose $z_0 = 1$. These two balls are related as $B(a) \subset B(a')$.

Henceforth, we use a as in (13) for describing a ball $B(a)$ and $\alpha_k \in \mathcal{I}_{p^k}$ is defined as the corresponding sum

$$\alpha_k := \sum_{i=0}^{k-1} a_i p^i. \quad (17)$$

For a fixed k , each $a \in \mathcal{I}_p^k$ has a unique integer sum α_k hence a ball $B(a)$ can be identified by its unique corresponding integer α_k .

Definition 17. (*Descriptor*) For $a := [a_0, a_1, \dots, a_{k-1}] \in \mathcal{I}_p^k$ the *descriptor* of ball $B(a)$ is α_k . The lower $\ell(a) = k - 1$, the smaller the range of possible values of α_k , and the larger the ball $B(a)$ whose descriptor is α_k .

Remark 18. From (17), the residue class corresponding to a ball $B(a)$ can be concisely expressed as $\alpha_k + p^k \mathbb{Z}_p$.

Definition 19. For a set $E \subset \mathbb{Z}_p$ and $x \in \mathbb{Z}_p$, define

$$d(x, E) := \inf_{z \in E} d(x, z).$$

Proposition 20. Let $a = [a_0, \dots, a_l] \in \mathbb{T}_p$ and $x \in \mathbb{Z}_p$ satisfy $x \notin B(a)$. Then

$$d(x, B(a)) = d(x, a).$$

Proof. Consider $x \notin B(a)$. Let $\tilde{a} := (a_0, \dots, a_l, 0, \dots)$, then $\tilde{a} \in B(a)$. For any $z \in B(a)$, we have $\ell(LCA(z, \tilde{a})) \geq \ell(a)$ because a is an ancestor of $\tilde{a}$ and z . Therefore

$$\begin{aligned} d(z, \tilde{a}) &= 1/p^{\ell(LCA(z, \tilde{a}))+1} && \text{(Proposition 11)} \\ &\leq 1/p^{\ell(a)+1} \\ &= r_a. && \text{(by (14))} \end{aligned}$$

Since every point of $B(a)$ may serve as its center (Remark 15), the assumption $x \notin B(a)$ implies that $d(x, z) > r_a$. Therefore $d(z, \tilde{a}) < d(x, z)$. By Proposition 2 it follows that $d(x, z) = d(x, \tilde{a})$ and therefore we have $\inf_{z \in B(a)} d(x, z) = d(x, \tilde{a})$. So

$$\begin{aligned} d(x, B(a)) &= \inf_{z \in B(a)} d(x, z) \\ &= d(x, \tilde{a}) \\ &= 1/p^{\ell(LCA(x, \tilde{a}))+1} && \text{(by Proposition 11)} \\ &= 1/p^{\ell(LCA(x, a))+1} \\ &= d(x, a) && \text{(by Definition 14).} \end{aligned}$$

□

3 p -adic concepts

In the Probably Approximately Correct (PAC) framework of learnability [30–32], a hypothesis is a binary classifier that can be used to infer some unknown concept from a finite sample.

By [3, Proposition 18.6] every open set in $\mathbb{Z}_p$ is a union of *disjoint* balls. This motivates the following definition of a p -adic concept.

Definition 21. (*p-adic concept*) A p -adic concept is a finite union of balls

$$U = \bigcup_{a \in A} B(a), \tag{18}$$

where A is a finite non-empty subset of distinct nodes of $\mathbb{T}_p$ and $U \neq \mathbb{Z}_p$.

Remark 22. While the above definition does not assume that the balls $B(a)$ are disjoint, every p -adic concept U admits a representation (18) for which $|A|$ is minimal among all finite representations of U as a union of balls. We call such a representation *reduced*. Equivalently, the balls in a reduced representation are precisely the maximal p -adic balls contained in U . In a reduced representation the balls are pairwise disjoint. (If two balls are not disjoint, then by Proposition 3 one is contained in the other, and removing the smaller ball gives a representation of U using fewer balls.) Moreover, a reduced representation cannot contain all p sibling nodes having the same parent because the union of the corresponding p balls is the ball corresponding to their parent, so replacing these p balls by the parent ball would give a representation of U using fewer balls.

Let us denote by $\mathcal{C}_n$ the class of p -adic concepts that consist of a union of up to n balls

$$\mathcal{C}_n := \left\{ U : U = \bigcup_{a \in A} B(a), 1 \leq |A| \leq n, U \neq \mathbb{Z}_p \right\}. \quad (19)$$

Next, we define binary classifiers on $\mathbb{Z}_p$ based on p -adic concepts.

From Definition 21, a p -adic concept is a union $U \subseteq \mathbb{Z}_p$ of balls,

$$U := U_A = \bigcup_{a \in A} B(a). \quad (20)$$

Corresponding to U we define a binary classifier $h : \mathbb{Z}_p \rightarrow \{-1, 1\}$,

$$h_U(x) := \begin{cases} 1, & x \in U \\ -1, & \text{otherwise.} \end{cases} \quad (21)$$

A classifier h acts as an indicator function for the union of balls associated with a p -concept where instead of 0 we use -1 to denote not being in U .

For a finite integer $n \geq 1$ denote the family of binary classifiers based on p -adic concepts by

$$\mathcal{H}_n = \{h_U : U \in \mathcal{C}_n\}. \quad (22)$$

4 Width

For $U \subseteq \mathbb{Z}_p$, denote by $U^c := \mathbb{Z}_p \setminus U$ the complement of U . Next, we extend the notion of width introduced in [15] to functions $h \in \mathcal{H}_n$.

Definition 23. (*Width*) For $h := h_U \in \mathcal{H}_n$, denote by $w_h : \mathbb{Z}_p \rightarrow \{p^{-k} : k \in \mathbb{Z}_{\geq 0}\} \cup \{0\}$ the *width* $w_h(x)$ of h at x and define it as follows,

$$w_h(x) := \begin{cases} d(x, U^c), & x \in U \\ d(x, U), & x \in U^c. \end{cases}$$

We assume throughout that the representation $U = \bigcup_{a \in A} B(a)$ is reduced in the sense of Remark 22. The width of h at x is evaluated as follows: if $x \in U$, there exists a node a such that $x \in B(a)$. We claim that $B(\pi(a)) \not\subseteq U$. If this were not true, then the balls in the reduced representation of U (including $B(a)$) that are contained in $B(\pi(a))$ could be replaced by the single ball $B(\pi(a))$, yielding a representation of U with fewer balls, contrary to Remark 22.

Hence $B(\pi(a)) \cap U^c \neq \emptyset$. The set U is a finite union of p -adic balls, each of which is closed (Proposition 4 and Proposition 6), therefore U^c is open. And $B(\pi(a))$ is open, therefore $B(\pi(a)) \cap U^c$ is a non-empty

open subset of $\mathbb{Z}_p$. By [3, Proposition 18.6], it therefore contains a p -adic ball $B(a')$, for some $a' \in \mathbb{T}_p$, such that

$$B(a') \subseteq B(\pi(a)) \cap U^{\mathbb{G}}.$$

Thus a' is a descendant of $\pi(a)$ but not a descendant of a (since $B(a) \subseteq U$). Consequently, for any $z \in B(a')$, $\text{LCA}(x, z) = \pi(a)$. Therefore

$$d(x, z) = \frac{1}{p^{\ell(\pi(a))+1}} = \frac{1}{p^{\ell(a)}}. \quad (23)$$

And for every $y \in U^{\mathbb{G}}$, since $B(a) \subseteq U$, we have $y \notin B(a)$. Hence $\ell(\text{LCA}(x, y)) < \ell(a)$, and therefore

$$d(x, y) = \frac{1}{p^{\ell(\text{LCA}(x, y))+1}} \geq \frac{1}{p^{\ell(a)}}. \quad (24)$$

Since $z \in U^{\mathbb{G}}$, it follows from (23) and (24) that

$$d(x, U^{\mathbb{G}}) = d(x, z).$$

Therefore, we have

$$\begin{aligned} w_h(x) &= d(x, z) \\ &= \frac{1}{p^{\ell(\text{LCA}(x, z))+1}} \end{aligned} \quad (25)$$

$$\begin{aligned} &= \frac{1}{p^{\ell(\pi(a))+1}} \\ &= d(x, \pi(a)) \end{aligned} \quad (26)$$

$$\begin{aligned} &= \frac{1}{p^{\ell(a)}} \end{aligned} \quad (27)$$

$$= pr_a \quad (28)$$

where (25) follows from Proposition 11, (26) follows from Definition 14 since $\pi(a)$ is an ancestor of x , (27) follows from $\ell(\pi(a)) + 1 = \ell(a)$ and (28) follows from (14).

If $x \notin U$, then by Definition 23,

$$\begin{aligned} w_h(x) &= d(x, U) \\ &= \inf_{z \in U} d(x, z) \quad (\text{by Definition 19}) \\ &= \min_{a \in A} \inf_{z \in B(a)} d(x, z) \\ &= \min_{a \in A} d(x, B(a)) \quad (\text{by Definition 19}) \\ &= \min_{a \in A} d(x, a) \quad (\text{by Proposition 20}) \\ &= \min_{a \in A} 1/p^{\ell(\text{LCA}(x, a))+1} \quad (\text{by Definition 14}) \\ &= 1/p^{\max_{a \in A} \ell(\text{LCA}(x, a))+1}. \end{aligned} \quad (29)$$

Proposition 24. Let $U = \bigcup_{a \in A} B(a)$ be a p -adic concept with reduced representation, and let $h := h_U \in \mathcal{H}_n$. Then, for every $x \in \mathbb{Z}_p$,

$$w_h(x) = \begin{cases} pr_a & \text{if } h(x) = 1, x \in B(a) \text{ for some } a \in A, \\ 1/p^{\max_{a \in A} \ell(\text{LCA}(x, a))+1} & \text{if } h(x) = -1, x \notin \bigcup_{a \in A} B(a). \end{cases} \quad (30)$$

Proof. The stated formula follows from the preceding derivation. $\square$

In case $h(x) = 1$ we can see from (30) that the *more shallow* (lower level ℓ) the LCA of x and a , where $B(a)$ is a ball that contains x , the *larger* the width $w_h(x)$ of h at a point x . Similarly, in case $h(x) = -1$, the more shallow the deepest LCA of x and a over all balls $B(a)$ in the union, where no such ball contains x , the *larger* the width $w_h(x)$ of h at x .

In other words, if $h(x) = 1$, the more shallow the node a whose corresponding ball contains x , the lower $\ell(a)$, the larger the radius r_a , and the larger the width $w_h(x)$ at x . In case $h(x) = -1$, the more shallow the LCA of the closest node $a \in A$ to x , the lower its level, the larger its ball $B(a)$, and the larger the width $w_h(x)$ at x .

In both cases, a larger width is associated with a larger ball, and a smaller range of possible values of descriptor α by Definition 17. By (7), this means having a *simpler* residue class because α is limited to a smaller set of values.

This shows a relationship between the width of a classifier and a number-theoretic representation α of the ball of a concept based on which this width value is obtained, which is summarized by the next statement.

Remark 25. For p -adic concepts, width is proportional to a number-theoretic representation of simplicity.

In later sections, we show that there is an advantage in learning classifiers with a large width on a sample of points. If a learning algorithm obtains a classifier that has a large width on a sample of points, in effect, it obtains a simpler p -adic concept.

Denote by ξ a sample of labeled examples, $\xi := \{[x^{(i)}, y_i]\}_{i=1}^m$, where $x^{(i)} \in \mathbb{Z}_p$ and $y_i \in \{-1, 1\}$, $1 \leq i \leq m$.

In Definition 23 the width function $w_h(x)$ is the p -adic distance from x to the nearest oppositely classified region by h thus it takes a non-negative value.

When evaluating the width of an example $[x, y] \in \mathbb{Z}_p \times \{-1, 1\}$ it is useful to take into account not only x but also y , hence we define a signed version of the width function next.

Definition 26. (*Signed width*) Define the *signed width* $f_h(x)$ as

$$f_h(x) := h(x)w_h(x). \quad (31)$$

Next, we define the sample-width.

Definition 27. (*Sample width*) The sample width of h at ξ is defined as

$$w_\xi(h) := \min_{1 \leq i \leq m} y_i f_h(x^{(i)}).$$

If h misclassifies an example, then $h(x^{(i)}) \neq y_i$, and hence $y_i f_h(x^{(i)}) < 0$, which makes the sample width $w_\xi(h)$ negative. Otherwise, h correctly classifies the sample and the sample width is non-negative.

As mentioned above, in a subsequent section we show that learning large width classifiers yields lower error bounds hence next we give some examples of classifiers that have a maximum width.

Example 28. Consider a sample of two examples $\xi = \{[x_+, 1], [x_-, -1]\}$. We wish to find a classifier h based on a concept U that consists of a single ball such that the sample width $w_\xi(h)$ is large. Consider the $LCA(x_+, x_-)$, then it has child nodes a_+ and a_- such that $x_+ \in B(a_+)$, a_- is an ancestor of x_- and $x_- \notin B(a_+)$. Let us consider the classifier h whose concept is $B(a_+)$. By (30), the width at x_+ is $w_h(x_+) = 1/p^{\ell(a_+)} = 1/p^{\ell(LCA(x_+, x_-))+1}$ and $w_h(x_-) = 1/p^{\ell(LCA(x_-, a_+))+1} = 1/p^{\ell(LCA(x_+, x_-))+1}$. Any other ball $B(a)$ that contains x_+ with $\ell(a) < \ell(a_+)$ must contain x_- and hence a classifier based on $B(a)$ as a concept, will misclassify it as 1. Any other ball $B(a)$ that contains x_+ and has a larger level $\ell(a) > \ell(a_+)$, has a lower width $w_h(x_+)$ and the same value for $w_h(x_-) = 1/p^{\ell(LCA(x_+, x_-))+1}$ as the ball $B(a_+)$. Therefore the classifier h based on a concept with a single ball $B(a_+)$ yields the maximum sample width $w_\xi(h) = 1/p^{\ell(LCA(x_+, x_-))+1}$ over all possible h based on a single ball.

Example 29. Let $m_+, m_- \geq 1$ be finite integers. Consider a sample of m_+ positive examples $\xi_+ = \{[x_i^+, 1]\}_{i=1}^{m_+}$ examples, and a sample of m_- negative examples $\xi_- = \{[x_j^-, -1]\}_{j=1}^{m_-}$ and denote by $\xi = \xi_+ \cup \xi_-$. We wish to find a classifier h based on a concept U which has some finite number of balls, is consistent with the sample, namely, classifies it correctly, and has maximal sample width $w_\xi(h)$. For each positive example x_i^+ define

$$r_i := \max \{ \ell(LCA(x_i^+, x_j^-)) : 1 \leq j \leq m_- \}$$

and denote by $\hat{x}_i^-$ a negative example for which $\ell(LCA(x_i^+, \hat{x}_i^-)) = r_i$. Any concept that correctly classifies the sample cannot have width at x_i^+ larger than $1/p^{r_i+1}$ since otherwise the corresponding ball that contains x_i^+ would also contain $\hat{x}_i^-$. It follows that the sample width of any classifier which is consistent with the sample ξ is no larger than $\mu := \min \{1/p^{r_i+1} : 1 \leq i \leq m_+\}$.

Let us now present a procedure that finds a classifier with a large sample width. For each positive example x_i^+ find a negative example $\hat{x}_i^-$ with which it has the deepest LCA, as above; the same negative example may be associated with different positive examples. Select x_*^+ with its associated $\hat{x}_*^-$, denoted by x_*^- , that has the most shallow LCA,

$$\ell(LCA(x_*^+, x_*^-)) = \min \{ \ell(LCA(x_i^+, \hat{x}_i^-)) : 1 \leq i \leq m_+ \}.$$

Without loss of generality, suppose this example is $x_1^+ = x_*^+$ and its associated negative example is $x_1^- = x_*^-$. Now, follow the steps of Example 28 to construct a ball $B(a_1^+)$ based on a child node a_1^+ of $LCA(x_1^+, x_1^-)$ such that $x_1^+ \in B(a_1^+)$ and $x_1^- \notin B(a_1^+)$. Note that no other negative example is contained in $B(a_1^+)$ as implied by definition of $\hat{x}_1^-$, which means x_1^- has the deepest LCA with x_1^+ . Let $U = B(a_1^+)$. If all positive examples are contained in U then halt. Otherwise, from the remaining positive examples x_i^+ , $2 \leq i \leq m_+$, select x_*^+ with its associated $\hat{x}_*^-$, denoted by x_*^- , that has the most shallow LCA,

$$\ell(LCA(x_*^+, x_*^-)) = \min \{ \ell(LCA(x_i^+, \hat{x}_i^-)) : 2 \leq i \leq m_+ \}.$$

Without loss of generality, suppose this example is $x_2^+ = x_*^+$ and its associated negative example is $x_2^- = x_*^-$. As above, repeat the steps of Example 28 to construct a ball $B(a_2^+)$ and let the concept $U = U \cup B(a_2^+)$. And so on, repeat this process until U contains all positive examples. The number of balls in the union U is no larger than m_+ hence U consists of a finite number of balls. The resulting concept U satisfies:

- every ball constructed in the procedure excludes all negative examples
- every negative example is at distance at least μ from U
- every positive example is either selected at some stage and then gets width $1/p^{r_i+1}$ or is already covered by an earlier ball and then has width at least μ

hence the maximal sample width over all consistent classifiers representable by a finite union of balls is μ and the classifier h_U realizes this maximum.

5 Dimensions

In the current section we deal with a general distance space $\mathcal{X}$ and with a distance function $\rho : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$. We assume $\rho(x, x) = 0$ for all $x \in \mathcal{X}$, ρ is non-negative, and the space is bounded (for any $x, x' \in \mathcal{X}$, $\rho(x, x') \leq R$, where R is a real positive number). We do not assume that the distance function ρ satisfies the standard metric requirements of identity of indiscernible, symmetry, and triangle inequality [33].

We recall some notions in combinatorics of sets [25].

Definition 30. (*Touching*) Let $\mathcal{C} = \{E_j\}_{j=1}^m$ be a collection of sets $E_j \subset \mathcal{X}$. We say that $\mathcal{C}$ is *touched* by $\mathcal{X}$ if for every set A_i of the form

$$A_i = \bigcap_{j=1}^m E_j^*$$

there exists a point $z_i \in \mathcal{X}$ such that $z_i \in A_i$, $1 \leq i \leq 2^m$, where E_j^* is either E_j or its complement $E_j^c := \mathcal{X} \setminus E_j$.

Denote by $2^{\mathcal{X}}$ the family of all subsets of $\mathcal{X}$.

Definition 31. (*Independence number [34]*) Let $\mathcal{Q} \subset 2^{\mathcal{X}}$ be a (possibly infinite) family of sets in $\mathcal{X}$. The independence number (IN) of $\mathcal{Q}$, denoted by $IN(\mathcal{Q})$, is the cardinality of the *largest* collection $\mathcal{C} \subset \mathcal{Q}$ touched by $\mathcal{X}$. If this largest collection is not finite or if its cardinality is an arbitrarily large finite number then we define $IN(\mathcal{Q}) := \infty$.

A *ball* of radius r centered at a point x in $\mathcal{X}$ is defined as

$$B_r(x) := \{z : \rho(z, x) \leq r\}.$$

Denote the family of all balls $\mathcal{B} \subset 2^{\mathcal{X}}$ in $\mathcal{X}$ by

$$\mathcal{B} := \{B_r(x) : x \in \mathcal{X}, r > 0\}. \quad (32)$$

Definition 32. (*Combinatorial dimension*) The *combinatorial dimension* of a distance space $\mathcal{X}$ denoted by $\dim(\mathcal{X})$ is defined to be $IN(\mathcal{B})$.

Lemma 33. $\dim(\mathbb{Z}_p) = 1$.

Proof. From Proposition 6, $\mathbb{Z}_p$ is an ultrametric space hence Proposition 3 implies that, for any pair of balls B_1, B_2 in $\mathbb{Z}_p$ exactly one of the following holds: (i) $B_1 \cap B_2 = \emptyset$, (ii) $B_1 \subseteq B_2$, or (iii) $B_2 \subseteq B_1$. In case (i) the region $B_1 \cap B_2$ is empty. In case (ii), $B_1 \cap B_2^c = \emptyset$. In case (iii), $B_1^c \cap B_2 = \emptyset$. Hence $\{B_1, B_2\}$ is not touched. Therefore no pair of balls is touched and therefore $\dim(\mathbb{Z}_p) = IN(\mathcal{B}) \leq 1$. On the other hand, any non-empty ball $B \subsetneq \mathbb{Z}_p$ is touched by $\mathbb{Z}_p$, since both B and B^c are non-empty. Hence $\dim(\mathbb{Z}_p) \geq 1$. $\square$

Remark 34. The proof of Lemma 33 uses only the property that balls in the underlying metric space are either nested or disjoint. Consequently, the same proof applies to any ultrametric space X with at least two points. In particular, it applies to leaf spaces of rooted trees equipped with an ultrametric. Hence, for every such space X , one has $\dim(X) = 1$.

Let $\mathbb{R}$ denote the set of real numbers. For $x \in \mathbb{R}$ denote the sign function by

$$\text{sgn}(x) := \begin{cases} -1 & \text{if } x \leq 0, \\ 1 & \text{if } x > 0. \end{cases}$$

For a real-valued function g , a set $S = \{x_i\}_{i=1}^m$ and a vector $r \in \mathbb{R}^m$, denote by

$$\text{sgn}(g|_S - r) := [\text{sgn}(g(x_1) - r_1), \dots, \text{sgn}(g(x_m) - r_m)].$$

Definition 35. (*Pseudo-dim*) For a class $\mathcal{G}$ of real-valued functions $g : \mathcal{X} \rightarrow \mathbb{R}$ the *Pseudo dimension* of $\mathcal{G}$ is defined as

$$Pdim(\mathcal{G}) := \max \{m \geq 1 : \exists r \in \mathbb{R}^m, \exists S \in \mathcal{X}^m, \{\text{sgn}(g|_S - r) : g \in \mathcal{G}\} = \{-1, 1\}^m\}. \quad (33)$$

If there exist a set $S = \{x_1, \dots, x_m\} \subseteq \mathcal{X}$ and a vector $r \in \mathbb{R}^m$ such that $\{\text{sgn}(g|_S - r) : g \in \mathcal{G}\} = \{-1, 1\}^m$, then we say that S is *shattered* by $\mathcal{G}$ (with respect to the threshold vector r).

For $g \in \mathcal{G}$ denote its max-norm on S by

$$\|g\|_S := \max_{x \in S} |g(x)|. \quad (34)$$

The ϵ -covering number of $\mathcal{G}$ with respect to the max-norm on S , denoted by $\mathcal{N}(\mathcal{G}, \epsilon, \|\cdot\|_S)$, is defined as the *minimum* integer k such that there exists an ϵ -cover of $\mathcal{G}$ (with respect to this norm) of size k .

Define the uniform ϵ -covering number of $\mathcal{G}$ as

$$\mathcal{N}_\infty(\mathcal{G}, \epsilon, m) := \max_{S \subseteq \mathcal{X}: |S|=m} \mathcal{N}(\mathcal{G}, \epsilon, \|\cdot\|_S) \quad (35)$$

and if the maximum does not exist then we let it be infinite.

Definition 36. For $z \in \mathcal{X}$ define $d_z : \mathcal{X} \rightarrow [0, R]$ as

$$d_z(x) := \rho(z, x), \quad (36)$$

and define the class $\mathcal{D}$ of distance functions by

$$\mathcal{D} := \{d_z : z \in \mathcal{X}\}. \quad (37)$$

From [25, Theorem 4], the class $\mathcal{D}$ has

$$Pdim(\mathcal{D}) = dim(\mathcal{X}), \quad (38)$$

where the combinatorial dimension $dim(\mathcal{X})$ is defined in Definition 32.

From [11, Theorem 12.2], for $m \geq Pdim(\mathcal{D})$, the covering number of the class $\mathcal{D}$ is bounded using its Pseudo-dimension as follows,

$$\mathcal{N}_\infty(\mathcal{D}, \gamma, m) \leq \left(\frac{emR}{\gamma Pdim(\mathcal{D})} \right)^{Pdim(\mathcal{D})}. \quad (39)$$

Let

$$\mathcal{D}_p := \{d_a : a \in \mathbb{T}_p\}$$

denote a class of distance functions (11) on $\mathbb{Z}_p$.

Proposition 37. We have,

$$Pdim(\mathcal{D}_p) = 1. \quad (40)$$

The proof is in Appendix A.

Substituting (40) for $Pdim(\mathcal{D})$ and (9) for R in (39) yields the bound

$$\mathcal{N}_\infty(\mathcal{D}_p, \gamma, m) \leq \frac{em}{\gamma}. \quad (41)$$

Remark 38. The class $\mathcal{C}_1 = \mathcal{B}_p \setminus \{\mathbb{Z}_p\}$ consists of nontrivial single-ball p -adic concepts and is the case $n = 1$ in (19). Moreover, for a node $a = [a_0, \dots, a_l] \in \mathbb{T}_p$, we may describe the ball $B(a)$ in terms of the distance function (11) as

$$B(a) = \{x : d_a(x) \leq r_a\},$$

where, from (14), $r_a = \frac{1}{p^{\ell(a)+1}}$. In the next section, we use this fact to define approximations of signed-width functions (Definition 26) of p -adic concepts.

6 $\mathcal{F}_n$

In this section we construct finite approximating families for the class $\mathcal{F}_n$ of signed-width functions associated with p -adic concepts consisting of unions of at most n balls. The purpose of this construction is to obtain explicit covering-number bounds for $\mathcal{F}_n$, which are needed in the next section to derive generalization bounds for large-width classification on $\mathbb{Z}_p$. The construction combines two ingredients developed earlier: the low complexity of the distance-function class $\mathcal{D}_p$, whose pseudo-dimension equals 1, and the hierarchical structure induced by p -adic balls on a finite sample. In particular, intersections of balls with a sample generate a dendrogram-like partition structure, and counting these admissible partitions yields a finite cover of $\mathcal{F}_n$ whose cardinality can be bounded explicitly.

From (26), (29) and (30), the signed width function (31) corresponding to a concept (20) is expressed in terms of functions d_a in $\mathcal{D}_p$, as follows

$$f_h(x) := f_A(x) = \begin{cases} d_{\pi(a)}(x) & \text{if } x \in B(a) \text{ for some } a \in A, \\ -\min_{a \in A} d_a(x) & \text{if } x \notin \bigcup_{a \in A} B(a), \end{cases} \quad (42)$$

where $A \subset \mathbb{T}_p$, $1 \leq |A| \leq n$, and $\bigcup_{a \in A} B(a)$ is a reduced representation of a p -adic concept in $\mathcal{C}_n$.

The class of signed width functions (31) is denoted by

$$\mathcal{F}_n = \{f_h : h \in \mathcal{H}_n\} \quad (43)$$

where $\mathcal{H}_n$ is defined in (22), or equivalently, $\mathcal{F}_n$ is the class of all functions f_A where $A \subset \mathbb{T}_p$, $1 \leq |A| \leq n$, and $\bigcup_{a \in A} B(a)$ is a reduced representation of a p -adic concept.

Fix a set $S = \{x_1, \dots, x_m\} \subset \mathbb{Z}_p$. Let

$$\hat{D}_\epsilon := \{\hat{d}_1, \dots, \hat{d}_{M_\epsilon}\}$$

be an ϵ -cover of $\mathcal{D}_p$ of minimal cardinality

$$M_\epsilon := \left| \hat{D}_\epsilon \right| \quad (44)$$

with respect to the norm (34).

From (35), we have

$$M_\epsilon = \mathcal{N}(\mathcal{D}_p, \epsilon, \|\cdot\|_S) \leq \mathcal{N}_\infty(\mathcal{D}_p, \epsilon, m). \quad (45)$$

Let

$$E = \{e_1, \dots, e_q\} \subset \mathbb{T}_p, \quad k = [k_1, \dots, k_q], \quad l = [l_1, \dots, l_q], \quad 1 \leq q \leq n, \quad (46)$$

with $k_i, l_i \in \{1, 2, \dots, M_\epsilon\}$, $1 \leq i \leq q$, such that the balls $B(e_i)$, $1 \leq i \leq q$, are disjoint. We define $\hat{f} : S \rightarrow \mathbb{R}$, where for $x \in S$,

$$\hat{f}_{E,k,l}(x) := \begin{cases} \hat{d}_{k_i}(x) & \text{if } x \in B(e_i) \cap S \text{ for some } e_i \in E \\ -\min_{1 \leq i \leq q} \hat{d}_{l_i}(x) & \text{if } x \in S \setminus \bigcup_{1 \leq i \leq q} B(e_i). \end{cases} \quad (47)$$

Next, we use (47) to construct an approximation for f_A over S .

Proposition 39. *Let $S = \{x_1, x_2, \dots, x_m\} \subset \mathbb{Z}_p$ and $A = \{a_1, \dots, a_q\} \subset \mathbb{T}_p$, $1 \leq q \leq n$. Let $f_A \in \mathcal{F}_n$. Define a selector $\sigma : \mathcal{D}_p \rightarrow \{1, 2, \dots, M_\epsilon\}$ where for $a_i \in A$,*

$$j_i := \sigma(d_{a_i}), \quad j_i^\pi := \sigma(d_{\pi(a_i)}) \quad (48)$$

and

$$\left\| d_{a_i} - \hat{d}_{j_i} \right\|_S \leq \epsilon, \quad \left\| d_{\pi(a_i)} - \hat{d}_{j_i^\pi} \right\|_S \leq \epsilon,$$

where $\hat{d}_{j_i}, \hat{d}_{j_i^\pi} \in \hat{D}_\epsilon$, $1 \leq i \leq q$. In (47), substitute

$$E = A, \quad k = j^\pi, \quad l = j \quad (49)$$

to obtain $\hat{f}_{A,j^\pi,j} : S \rightarrow \mathbb{R}$. Then

$$\left\| f_A - \hat{f}_{A,j^\pi,j} \right\|_S \leq \epsilon.$$

Proof. Fix $x \in S$ and consider the case that $x \in B(a_i)$, for some $a_i \in A$. Then

$$\begin{aligned} \left| f_A(x) - \hat{f}_{A,j^\pi,j}(x) \right| &= \left| d_{\pi(a_i)}(x) - \hat{d}_{\sigma(d_{\pi(a_i)})}(x) \right| \\ &= \left| d_{\pi(a_i)}(x) - \hat{d}_{j_i^\pi}(x) \right| \\ &\leq \left\| d_{\pi(a_i)} - \hat{d}_{j_i^\pi} \right\|_S \\ &\leq \epsilon. \end{aligned} \quad (50)$$

Now, consider the case $x \in S \setminus \bigcup_{a \in A} B(a)$. Then

$$\begin{aligned} \left| f_A(x) - \hat{f}_{A,j^\pi,j}(x) \right| &= \left| -\min_{1 \leq i \leq q} d_{a_i}(x) - \left(-\min_{1 \leq i \leq q} \hat{d}_{\sigma(d_{a_i})}(x) \right) \right| \\ &= \left| \min_{1 \leq i \leq q} \hat{d}_{j_i}(x) - \min_{1 \leq i \leq q} d_{a_i}(x) \right|. \end{aligned} \quad (51)$$

Let s be an index for which

$$d_{a_s}(x) = \min_{1 \leq i \leq q} d_{a_i}(x).$$

If there is more than one such index, choose any one of them. Then

$$\begin{aligned} \min_{1 \leq i \leq q} \hat{d}_{j_i}(x) - \min_{1 \leq i \leq q} d_{a_i}(x) &= \min_{1 \leq i \leq q} \hat{d}_{j_i}(x) - d_{a_s}(x) \\ &\leq \hat{d}_{j_s}(x) - d_{a_s}(x) \\ &\leq \max_{1 \leq i \leq q} \left| \hat{d}_{j_i}(x) - d_{a_i}(x) \right|. \end{aligned} \quad (52)$$

Similarly, let t be an index for which

$$\hat{d}_{j_t}(x) = \min_{1 \leq i \leq q} \hat{d}_{j_i}(x).$$

Then

$$\begin{aligned} \min_{1 \leq i \leq q} d_{a_i}(x) - \min_{1 \leq i \leq q} \hat{d}_{j_i}(x) &= \min_{1 \leq i \leq q} d_{a_i}(x) - \hat{d}_{j_t}(x) \\ &\leq d_{a_t}(x) - \hat{d}_{j_t}(x) \\ &\leq \max_{1 \leq i \leq q} \left| d_{a_i}(x) - \hat{d}_{j_i}(x) \right|. \end{aligned} \quad (53)$$

For any $x \in S$ we have $\left| d_{a_i}(x) - \hat{d}_{j_i}(x) \right| \leq \left\| d_{a_i} - \hat{d}_{j_i} \right\|_S$ and since $\hat{D}_\epsilon$ covers $\mathcal{D}_p$ then for any $i \in \{1, \dots, q\}$, $\left\| d_{a_i} - \hat{d}_{j_i} \right\|_S \leq \epsilon$. It follows that the left-hand sides of (52) and (53) are each bounded from above by ϵ , and therefore the absolute value in (51) is also bounded from above by ϵ . With (50), it follows that for any $x \in S$, $\left| f_A(x) - \hat{f}_{A,j^\pi,j}(x) \right| \leq \epsilon$ therefore $\left\| f_A - \hat{f}_{A,j^\pi,j} \right\|_S \leq \epsilon$. $\square$

Two-step construction of $\hat{f}$: Fix a sample $S = \{x_1, \dots, x_m\} \subset \mathbb{Z}_p$. In constructing $\hat{f} : S \rightarrow \mathbb{R}$ according to (47), the value of $\hat{f}$ on S can be obtained through two steps:

- define disjoint sets $R_0, R_1, \dots, R_L \subseteq S$, whose non-empty members form a partition

$$\Pi_L(S) := \{R_j : 0 \leq j \leq L, R_j \neq \emptyset\}$$

of S , where, for $1 \leq j \leq L \leq q \leq n$, each R_j is of the form

$$R_j = S \cap B(e_i) \neq \emptyset \quad (54)$$

for some $e_i \in E$, and

$$R_0 = S \setminus \bigcup_{1 \leq i \leq q} B(e_i) \quad (55)$$

may be empty.

- choose the index vectors k and l that determine the values of $\hat{f}$ on the non-empty sets R_j , $0 \leq j \leq L$.

According to (47), for each $x \in S$, the value of $\hat{f}(x)$ depends on whether x is in R_0 or in R_j , $1 \leq j \leq L$, and is determined by functions in the minimal ϵ -cover $\hat{D}_\epsilon$ of $\mathcal{D}_p$, whose index belongs to the set

$$K := \{1, 2, \dots, M_\epsilon\} \quad (56)$$

where M_ϵ is defined in (44). Hence, a function $\hat{f}_{E,k,l}$ corresponds to a partition together with choices of the index vectors k and l , which determine its values on the parts of this partition.

Since there are a finite number of partitions $\Pi_L(S)$ of S , and a finite number of choices of the index vectors k and l that determine $\hat{f}$ on these parts, there are a finite number of mappings $\hat{f}_{E,k,l}$ defined in (47), despite the fact that there are infinitely many subsets $E \subset \mathbb{T}_p$.

Recall that a partition of S is a collection of pairwise disjoint, non-empty subsets of S , called *blocks*, whose union is S . Before stating and proving the next result, we introduce notation for the blocks in the dendrogram of a finite sample S .

Following [35, Sections 3.1 and 3.3.2], let θ_S denote the dendrogram associated with the finite ultrametric space (S, d) . At each scale $r \geq 0$, $\theta_S(r)$ is the partition of S into the equivalence classes of the relation

$$x \sim_r y \iff d(x, y) \leq r, \quad x, y \in S.$$

Denote by

$$\mathcal{H}_S := \bigcup_{r \geq 0} \theta_S(r)$$

the collection of all distinct blocks that occur in the dendrogram as r varies.

Lemma 40. *Let $S = \{x_1, \dots, x_m\} \subset \mathbb{Z}_p$. The number of distinct functions $\hat{f}_{E,k,l} : S \rightarrow \mathbb{R}$ of the form (47) is no more than*

$$M_\epsilon^n \sum_{L=0}^n \binom{2m-1}{L} M_\epsilon^L. \quad (57)$$

Proof. The proof combines two observations. First, every non-empty intersection of S with a p -adic ball is a block in $\mathcal{H}_S$, and $\mathcal{H}_S$ contains at most $2m - 1$ distinct blocks. Hence, for each $0 \leq L \leq q \leq n$, there are at most $\binom{2m-1}{L}$ choices of L blocks, and therefore at most this many partitions of S . Second, once such a partition is fixed, the choices of the discretization indices yield at most M_ϵ^{n+L} functions. Summing over L gives (57). The complete argument is given in Appendix B. $\square$

Given a sample $S \subset \mathbb{Z}_p$ and $f := f_A \in \mathcal{F}_n$, the two-step construction on the preceding page gives an ϵ -approximation $\hat{f}_{A,j^\pi,j}$ of f ,

$$\left\| f_A - \hat{f}_{A,j^\pi,j} \right\|_S \leq \epsilon. \quad (58)$$

Lemma 40 implies that the number of functions $\hat{f}_{A,j^\pi,j}$ is bounded from above by (57). Hence, with respect to the max-norm on S , there exists a finite ϵ -cover of $\mathcal{F}_n$, denoted by

$$\hat{F}_\epsilon := \left\{ \hat{f}_1, \hat{f}_2, \dots, \hat{f}_{N_\epsilon} \right\}. \quad (59)$$

Lemma 41. (Covering number of $\mathcal{F}_n$) Let $S \subset \mathbb{Z}_p$ be of cardinality $|S| = m$. Let M_ϵ be the covering number of $\mathcal{D}_p$ with respect to the max-norm on S (34). If $m \geq \lceil (n+1)/2 \rceil$ then the covering number $\mathcal{N}(\mathcal{F}_n, \epsilon, \|\cdot\|_S)$ of $\mathcal{F}_n$ with respect to the max-norm on S is no larger than

$$\mathcal{N}(\mathcal{F}_n, \epsilon, \|\cdot\|_S) \leq N_\epsilon := \left(\frac{eM_\epsilon^2(2m-1)}{n} \right)^n. \quad (60)$$

Proof. Fix a set $S \subset \mathbb{Z}_p$ of cardinality m . From Proposition 39, for any $f \in \mathcal{F}_n$ there exists an ϵ -approximation $\hat{f} \in \hat{F}_\epsilon$ of f with respect to the $\|\cdot\|_S$ norm, where $\hat{F}_\epsilon$ is defined in (59). Therefore $\hat{F}_\epsilon$ is an ϵ -cover of $\mathcal{F}_n$ with respect to this norm and its cardinality is bounded from above by

$$\begin{aligned} M_\epsilon^n \sum_{k=0}^n \binom{2m-1}{k} M_\epsilon^k &\leq M_\epsilon^{2n} \sum_{k=0}^n \binom{2m-1}{k} \\ &\leq M_\epsilon^{2n} \left(\frac{e(2m-1)}{n} \right)^n \\ &= \left(\frac{eM_\epsilon^2(2m-1)}{n} \right)^n \end{aligned} \quad (61)$$

where (61) holds if $m \geq \lceil (n+1)/2 \rceil$. □

7 Learning

Let P be any probability measure on some appropriate σ -algebra over the set $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$ where $\mathcal{Y} := \{-1, 1\}$. Functions on $\mathcal{Z}$ are assumed to be measurable without explicit reference. A sample is a finite set of labeled points $\{z_i\}_{i=1}^m$, $z_i = (x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, $1 \leq i \leq m$. The space of all samples of size m is denoted by $\mathcal{Z}^m$. A *random* sample $\xi := \{(X_i, Y_i)\}_{i=1}^m \subset \mathcal{Z}$ is drawn according to the joint (product) probability measure P^m which is defined on $\mathcal{Z}^m$, that is, ξ consists of m i.i.d. random pairs (X_i, Y_i) , $1 \leq i \leq m$, each drawn according to P .

We consider the problem of learning binary classification using the class $\mathcal{H}_n$ of classifiers based on p -adic concepts on $\mathbb{Z}_p$. By learning we mean the process of finding a good classifier h_U based on a p -adic concept U which has a low probability of misclassifying a randomly drawn labeled point $(X, Y) \in \mathcal{Z}$. This misclassification error is referred to as the *generalization error* of h_U and is defined as the following probability

$$P(h_U(X) \neq Y). \quad (62)$$

Our aim in this section is to obtain an upper bound on the generalization error of any h_U in $\mathcal{H}_n$.

A binary classifier $h := h_U$ can be expressed in terms of a signed width function (42) as follows

$$h(x) = \text{sgn}(f(x)) \quad (63)$$

where $f := f_A$, and A is the set of nodes in the definition (20) of concept U .

We choose to represent the classifier in terms of a real-valued function as in (63) because the error can then be bounded by a quantity which involves the covering number of the class $\mathcal{F}_n$ (43) which is bounded from above by an expression that depends on the Pseudo-dimension of $\mathcal{D}_p$. While a more direct approach to the analysis of learning $\mathcal{H}_n$ would involve the VC-dimension of $\mathcal{H}_n$ instead of the Pseudo-dimension of $\mathcal{F}_n$, this would leave us with an error bound that does not depend on the width parameter γ and such bounds are less tight.

We can express the generalization error of h in terms of f_A as follows,

$$\begin{aligned} P(h(X) \neq Y) &= P(Yf(X) < 0) + P(Y = 1, f(X) = 0) \\ &\leq P(Yf(X) \leq 0). \end{aligned} \quad (64)$$

Thus learning $\mathcal{H}_n$ amounts to learning the class $\mathcal{F}_n$, that is, the aim is to find an f for which (64) is low and this ensures that the error of h is low.

Formally, under the extension of the PAC model of learnability to real-valued functions [36] the learning problem is as follows: let $\mathcal{Z} := \mathbb{Z}_p \times \{-1, 1\}$. We wish to learn a binary classifier h by finding an f from the class $\mathcal{F}_n$ using a randomly drawn training sample $\xi := \{(X_i, Y_i)\}_{i=1}^m \subset \mathcal{Z}$, such that f has a small probability value (64) and therefore its corresponding h has a small error. This value (64) cannot be measured because the probability distribution P is assumed to be unknown so we need an empirical estimate. We follow the notion of large-margin learning [11, Ch. 13] and for $\gamma \in (0, \text{diam}(\mathbb{Z}_p)]$ we define the following empirical quantity,

$$P_m(Yf(X) \leq \gamma) := \frac{1}{m} \sum_{i=1}^m \mathbb{I}\{Y_i f(X_i) \leq \gamma\} \quad (65)$$

which we refer to as the *empirical γ -margin error* of f . The next result shows that given any γ in the range of possible margin values, with high confidence, uniformly over all $h \in \mathcal{H}_n$ the generalization error of h is not much greater than the empirical γ -margin error of its corresponding f .

Lemma 42. *Let $\mathcal{H}_n$ be a class of binary classifiers $h(x) = \text{sgn}(f(x))$ where*

$$f(x) := f_A(x) = \begin{cases} d_{\pi(a)}(x) & \text{if } x \in B(a) \text{ for some } a \in A, \\ -\min_{a \in A} d_a(x) & \text{if } x \notin \bigcup_{a \in A} B(a), \end{cases}$$

corresponds to a p -adic concept $U_A = \bigcup_{a \in A} B(a)$ with $A \subset \mathbb{T}_p$, $1 \leq |A| \leq n$. Denote the class of all such functions f by $\mathcal{F}_n$. Let P be any probability measure on $\mathcal{Z} := \mathbb{Z}_p \times \{-1, 1\}$ and let $\delta \in (0, 1]$ be a confidence parameter. For any integer $m \geq \lceil (n+1)/2 \rceil$ and $\gamma \in (0, 1]$ the following holds with P^m -probability at least $1 - \delta$ for a random training sample $\xi \in \mathcal{Z}^m$ drawn i.i.d. according to P^m : for all $f \in \mathcal{F}_n$,

$$P(h(X) \neq Y) \leq P_m(Yf(X) \leq \gamma) + \sqrt{\frac{8}{m} \left(n \ln \left(\frac{16e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{2}{\delta} \right) \right)}. \quad (66)$$

Proof. Define a margin function

$$\mu(f(x), y) := yf(x). \quad (67)$$

In [11], Theorem 10.1 is stated for a class of real valued functions that map a domain into the unit interval, with a different margin function (Definition 9.1, [11]) and binary classification values $\mathcal{Y} = \{0, 1\}$. The proof of that theorem holds for our setup by substituting $\mathcal{F}_n$, the margin function (67) and binary values $y \in \{-1, 1\}$, for the corresponding three quantities in their setup (the definition of empirical γ -margin error is the same in both setups). Therefore, the theorem yields the following bound,

$$P^m(\exists f \in \mathcal{F}_n, P(Yf(X) \leq 0) > P_m(Yf(X) \leq \gamma) + \epsilon) \leq 2\mathcal{N}_\infty(\mathcal{F}_n, \gamma/2, 2m) \exp(-\epsilon^2 m/8). \quad (68)$$

From Lemma 41, the right hand of (60) holds for any $S \subset \mathbb{Z}_p$ of cardinality m , hence it follows that the uniform covering number (35) is bounded as

$$\begin{aligned} \mathcal{N}_\infty(\mathcal{F}_n, \gamma/2, 2m) &\leq M_{\gamma/2}^{2n} \left(\frac{e(4m-1)}{n} \right)^n \\ &\leq \left(\frac{4em}{\gamma} \right)^{2n} \left(\frac{e(4m-1)}{n} \right)^n \end{aligned} \quad (69)$$

where (69) follows from (41) and (45).

This implies that the right hand side of (68) is bounded from above by the expression

$$2 \left(\frac{4em}{\gamma} \right)^{2n} \left(\frac{e(4m-1)}{n} \right)^n \exp(-\epsilon^2 m/8)$$

which is simplified to

$$2 \left(\frac{16e^3 m^2 (4m-1)}{\gamma^2 n} \right)^n \exp(-\epsilon^2 m/8).$$

Setting this to be δ and solving for ϵ yields (66). $\square$

Next is the main result of this section, which states that with a large confidence, uniformly over all possible values of γ and all classifiers $h \in \mathcal{H}_n$, the generalization error of h is not much greater than the empirical γ -margin error of its corresponding f .

Theorem 43. *With the same setup as in Lemma 42, let P be any probability measure on $\mathbb{Z}_p \times \{-1, 1\}$ and let $\delta \in (0, 1]$ be a confidence parameter. For any integer $m \geq \lceil (n+1)/2 \rceil$ the following holds with P^m -probability at least $1 - \delta$ for a random training sample $\xi \in \mathcal{Z}^m$ drawn i.i.d. according to P^m : for all $f \in \mathcal{F}_n$ and all $\gamma \in (0, 1]$,*

$$P(h(X) \neq Y) \leq P_m(Yf(X) \leq \gamma) + \sqrt{\frac{8}{m} \left(n \ln \left(\frac{64e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{4}{\gamma \delta} \right) \right)}. \quad (70)$$

Note that in Theorem 43, γ is not specified in advance of drawing the sample ξ . In particular, if h is consistent with the sample, then the supremum of the values of γ for which the empirical γ -margin error on ξ is zero is precisely the sample width of h on ξ .

Proof outline. For a fixed margin parameter, Lemma 42 provides the required deviation bound. To make the conclusion uniform in $\gamma \in (0, 1]$, partition the margin range into the dyadic intervals

$$\Gamma_l = (2^{-l}, 2^{-(l-1)}], \quad l \geq 1.$$

Monotonicity of the empirical margin event and of the deviation term allows every bad event with $\gamma \in \Gamma_l$ to be contained in a single bad event at the endpoint $\gamma_l = 2^{-l}$. Applying Lemma 42 with confidence level $\gamma_l \delta$ and summing the resulting probabilities over l gives a total failure probability at most

$$\delta \sum_{l=1}^{\infty} 2^{-l} = \delta.$$

Substitution of the corresponding deviation term yields (70). The complete argument is given in Appendix C. $\square$

In the bound (70), as γ increases the empirical γ -margin error of f increases while the square root term decreases. Thus there is a tradeoff between the two terms of the sum on the right side of (70). Theorem 43 can be used as a basis for ‘model selection’ [11, Ch. 15] in order to choose a classifier $h \in \mathcal{H}_n$ which minimizes this sum with respect to γ . Its generalization error is bounded from above by an expression of the form

$$\inf_{\gamma \in (0,1]} \inf_{f \in \mathcal{F}_n} \left(P_m(Yf(X) \leq \gamma) + \sqrt{\frac{8}{m} \left(n \ln \left(\frac{64e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{4}{\gamma \delta} \right) \right)} \right).$$

8 Conclusions

This paper studies large-width classification on the ultrametric domain of p -adic integers $\mathbb{Z}_p$. The hypothesis class is defined by finite unions of p -adic balls, and the associated signed-width function represents each classifier as the sign of a real-valued function. This places the analysis within the framework of margin-based learning [10, 11] while retaining the explicit hierarchical geometry of the p -adic setting.

A central feature of the analysis is the strong structural regularity of balls in $\mathbb{Z}_p$: any two balls are either nested or disjoint. From this we obtain $\dim(\mathbb{Z}_p) = 1$ and, consequently, pseudo-dimension 1 for the associated class of distance functions. This leads to covering-number bounds for the class of signed-width functions and, in turn, to non-asymptotic generalization bounds for learning concepts that are unions of at most n balls. The bounds hold uniformly over the width parameter $\gamma \in (0, 1]$, so γ may be selected after observing a sample of finite size m . They quantify the tradeoff between empirical width error and a complexity term depending on n , m , γ , and the confidence parameter δ .

The analysis also gives a concrete interpretation of width in the p -adic setting. Since a ball is determined by a finite prefix in the p -adic expansion [27], larger width corresponds to a coarser ball and therefore to a simpler residue-class description of the concept. Thus, maximizing sample width favors classifiers that are simpler in a natural number-theoretic sense.

More broadly, the results suggest that ultrametric geometry can support a tractable and informative learning theory for hierarchical domains. In this perspective, the space $\mathbb{Z}_p$ serves not merely as a specialized number-theoretic object, but as a mathematically transparent instantiation of a more general framework for classification and generalization on hierarchical ultrametric spaces. Possible directions for further work include algorithmic aspects of learning over $\mathbb{Z}_p$, multi-category versions of the model, extensions to other ultrametric domains, and empirical evaluation on genuinely hierarchical data. In particular, it would be of interest to compare the observed generalization error of learning algorithms for p -adic concepts with the non-asymptotic bounds derived in this paper, thereby assessing the practical tightness and predictive value of the theoretical guarantees.

A Proof of Proposition 37

To distinguish the two types of distance functions, throughout the following argument d_a , with $a \in \mathbb{T}_p$, denotes the node-distance function in $\mathcal{D}_p$, whereas d_z , with $z \in \mathbb{Z}_p$, denotes the point-distance function in $\mathcal{D}$, where in (37) we let $\mathcal{X} = \mathbb{Z}_p$.

We first show that

$$\text{Pdim}(\mathcal{D}_p) \geq 1.$$

Fix $x \in \mathbb{Z}_p$. Let $a \in \mathbb{T}_p$ be a non-root ancestor of x and let b be a sibling of a . Then

$$d_a(x) < d_b(x).$$

Choose the threshold $r = d_a(x)$. Then

$$\text{sgn}(d_a(x) - r) = -1$$

and

$$\text{sgn}(d_b(x) - r) = 1.$$

Hence $\{x\}$ is shattered by $\mathcal{D}_p$, and therefore

$$\text{Pdim}(\mathcal{D}_p) \geq 1.$$

We next show that

$$\text{Pdim}(\mathcal{D}_p) < 2.$$

Suppose, to the contrary, that there exist distinct $x, y \in \mathbb{Z}_p$ and a threshold vector $r = [r_1, r_2] \in \mathbb{R}^2$ such that $\{x, y\}$ is shattered by $\mathcal{D}_p$. Thus there exist four node-distance functions

$$d_a, \quad d_b, \quad d_c, \quad d_e \in \mathcal{D}_p$$

that realize the four sign vectors in $\{-1, 1\}^2$ on $\{x, y\}$ with respect to r , where $a, b, c, e \in \mathbb{T}_p$.

We show that each of these four node-distance functions can be replaced by a point-distance function in $\mathcal{D}$ having the same sign vector on $\{x, y\}$. It would then follow that $\mathcal{D}$ shatters $\{x, y\}$, contradicting $\text{Pdim}(\mathcal{D}) = 1$.

Consider any one of the four node-distance functions, say d_v , where $v \in \{a, b, c, e\}$, and let $B(v)$ have radius r_v . For every $u \in B(v)$,

$$d_v(u) = r_v.$$

Moreover, if $u \notin B(v)$ and $z \in B(v)$, then

$$\text{LCA}(u, z) = \text{LCA}(u, v),$$

and hence

$$d_z(u) = d_v(u). \tag{71}$$

If $u \in B(v)$ and $z \in B(v)$, then

$$d_z(u) \leq r_v = d_v(u). \tag{72}$$

We consider the four possible positions of x and y relative to $B(v)$.

First, suppose that

$$x \notin B(v), \quad y \notin B(v).$$

Choose any $z \in B(v)$. By (71),

$$d_z(x) = d_v(x), \quad d_z(y) = d_v(y),$$

so d_z has the same sign vector on $\{x, y\}$ as d_v .

Next, suppose that

$$x \in B(v), \quad y \notin B(v).$$

If d_v assigns the negative sign to x , then

$$r_v = d_v(x) \leq r_1.$$

Choose any $z \in B(v)$. By (72),

$$d_z(x) \leq r_v \leq r_1,$$

while, by (71),

$$d_z(y) = d_v(y).$$

Thus d_z has the same sign vector as d_v .

If d_v assigns the positive sign to x , then

$$r_v = d_v(x) > r_1.$$

Choose $z \in B(v)$ whose path in $\mathbb{T}_p$ passes through a child of v different from the child containing x . Such a child exists since $p \geq 2$. Then

$$\text{LCA}(x, z) = v,$$

and hence

$$d_z(x) = r_v = d_v(x) > r_1.$$

Since $y \notin B(v)$,

$$d_z(y) = d_v(y).$$

Therefore d_z again has the same sign vector as d_v .

The case

$$y \in B(v), \quad x \notin B(v)$$

is symmetric.

It remains to consider the case

$$x, y \in B(v).$$

Then

$$d_v(x) = d_v(y) = r_v.$$

If d_v has sign vector $(-1, -1)$, then

$$r_v \leq r_1, \quad r_v \leq r_2.$$

For any $z \in B(v)$,

$$d_z(x) \leq r_v \leq r_1, \quad d_z(y) \leq r_v \leq r_2,$$

so d_z also has sign vector $(-1, -1)$.

If d_v has sign vector $(1, -1)$, then

$$r_v > r_1, \quad r_v \leq r_2.$$

Choose $z \in B(v)$ whose path passes through a child of v different from the child containing x . Then

$$d_z(x) = r_v > r_1,$$

whereas

$$d_z(y) \leq r_v \leq r_2.$$

Thus d_z has sign vector $(1, -1)$. The case $(-1, 1)$ is symmetric.

Finally, suppose that d_v has sign vector $(1, 1)$. Then

$$r_v > r_1, \quad r_v > r_2.$$

If x and y lie in the same child of v , choose z in a different child of v . Then

$$d_z(x) = d_z(y) = r_v,$$

and hence d_z also has sign vector $(1, 1)$.

If x and y lie in different children of v and $p \geq 3$, choose z in a third child of v . Again,

$$d_z(x) = d_z(y) = r_v,$$

so d_z has sign vector $(1, 1)$.

It remains only to consider the case $p = 2$ in which x and y lie in the two different children of v . If $B(v) \neq \mathbb{Z}_p$, choose $z \notin B(v)$. Since $x, y \in B(v)$ while $z \notin B(v)$,

$$\text{LCA}(z, x) = \text{LCA}(z, y) = \text{LCA}(z, v),$$

and this node is a strict ancestor of v . Hence

$$\ell(\text{LCA}(z, x)) = \ell(\text{LCA}(z, y)) < \ell(v).$$

Therefore,

$$d_z(x) = \frac{1}{p^{\ell(\text{LCA}(z, x))+1}} > \frac{1}{p^{\ell(v)+1}} = r_v,$$

and similarly,

$$d_z(y) > r_v.$$

The only remaining possibility is that $p = 2$, $B(v) = \mathbb{Z}_p$, and x and y lie in the two different children of v . In this case $r_v = 1$. Since d_v has sign vector $(1, 1)$,

$$r_1 < 1, \quad r_2 < 1.$$

However, since $\{x, y\}$ is shattered by $\mathcal{D}_p$, there must also exist a node-distance function $d_w \in \mathcal{D}_p$ having sign vector $(-1, -1)$, and hence

$$d_w(x) \leq r_1 < 1, \quad d_w(y) \leq r_2 < 1.$$

This is impossible. Indeed, if w lies below the child containing x , then

$$d_w(y) = 1,$$

whereas if w lies below the child containing y , then

$$d_w(x) = 1.$$

For the root itself both values are equal to 1. Thus no $d_w \in \mathcal{D}_p$ can have sign vector $(-1, -1)$, a contradiction.

Consequently, each of the four sign vectors realized on $\{x, y\}$ by $\mathcal{D}_p$ can also be realized by a point-distance function in $\mathcal{D}$. Hence $\mathcal{D}$ shatters $\{x, y\}$, contradicting

$$\text{Pdim}(\mathcal{D}) = 1.$$

Therefore

$$\text{Pdim}(\mathcal{D}_p) < 2.$$

Together with $\text{Pdim}(\mathcal{D}_p) \geq 1$, it follows that

$$\text{Pdim}(\mathcal{D}_p) = 1. \tag{73}$$

B Proof of Lemma 40

Since a function $\hat{f}$ corresponds to a partition of S and choices of indices that determine its values on the parts of the partition, to bound from above the number of functions $\hat{f}$ we start by obtaining an upper bound on the number of partitions $\Pi_L(S)$ of S .

For $L \geq 1$, the non-empty parts R_i , $1 \leq i \leq L$, are of the form (54); that is,

$$R_i = S \cap B(e_i), \quad e_i \in E, \quad 1 \leq i \leq L.$$

The remaining set

$$R_0 = S \setminus \bigcup_{1 \leq i \leq L} B(e_i)$$

is an additional part of the partition when it is non-empty. If $L = 0$, the partition consists of the single part $R_0 = S$.

To bound the number of partitions $\Pi_L(S)$, it suffices to bound the number of possible combinations of $L \leq q$ non-empty subsets of S picked by p -adic balls $B \in \mathcal{B}_p$. Any elements of S that are not picked by these balls form the remaining part R_0 . Thus, each partition $\Pi_L(S)$ is determined by a combination of $L \leq q$ such subsets of S .

Next, we show that a single p -adic ball picks out a block of the dendrogram.

Claim 44. Let $B_r \in \mathcal{B}_p$ be a ball of radius r . If $B_r \cap S \neq \emptyset$, then $B_r \cap S$ is a block of $\theta_S(r)$, and hence $B_r \cap S \in \mathcal{H}_S$.

Proof. Choose $x \in B_r \cap S$. By Remark 15, every point of a p -adic ball may serve as its center. Therefore

$$B_r \cap S = \{y \in S : d(x, y) \leq r\}.$$

By definition of $\theta_S(r)$, the set on the right is exactly the equivalence class containing x under $\sim_r$, and hence is a block of $\theta_S(r)$. $\square$

It remains to bound the number of distinct blocks in $\mathcal{H}_S$. Since the partitions of the dendrogram are nested [35, Section 3.1], $\mathcal{H}_S$ satisfies the following *nesting property*: for any $A, B \in \mathcal{H}_S$, either $A \cap B = \emptyset$, $A \subseteq B$, or $B \subseteq A$. Indeed, if $A \in \theta_S(r)$ and $B \in \theta_S(s)$ with $r \leq s$, then $\theta_S(r)$ refines $\theta_S(s)$, so either $A \cap B = \emptyset$ or $A \subseteq B$. Moreover, $\mathcal{H}_S$ contains every singleton $\{x\}$, $x \in S$, as a block of $\theta_S(0)$, and it contains S itself at every scale r greater than or equal to $\text{diam}(S)$.

We claim that a family $\mathcal{L}$ of non-empty subsets of an m -element set with the above property, containing all singleton sets and the whole set, has cardinality at most $2m - 1$. We prove this by strong induction on m . For $m = 1$, we have $|\mathcal{L}| = 1 = 2(1) - 1$.

Let $m \geq 2$, and assume that the assertion holds for every set of cardinality less than m . Let $A_1, \dots, A_k$ be the maximal members of $\mathcal{L}$ under inclusion, excluding the whole set itself. The sets $A_1, \dots, A_k$ are pairwise disjoint and they cover the whole set, because every singleton belongs to $\mathcal{L}$ and is contained in one of the maximal members $A_1, \dots, A_k$. Since no A_j equals the whole set, we must have $k \geq 2$.

For each $j \in \{1, \dots, k\}$, let $\mathcal{L}_j := \{A \in \mathcal{L} : A \subseteq A_j\}$. Since A_j is not the whole set, $|A_j| < m$. The family $\mathcal{L}_j$ satisfies the nesting property on A_j and contains all singleton subsets of A_j as well as A_j itself. Hence, by the induction hypothesis applied to A_j , $|\mathcal{L}_j| \leq 2|A_j| - 1$. Every member of $\mathcal{L}$ other than the whole set belongs to exactly one of the families $\mathcal{L}_j$. Therefore,

$$|\mathcal{L}| = 1 + \sum_{j=1}^k |\mathcal{L}_j| \leq 1 + \sum_{j=1}^k (2|A_j| - 1) = 1 + 2m - k \leq 2m - 1. \quad (74)$$

Applying (74) to $\mathcal{H}_S$ we obtain

$$|\mathcal{H}_S| \leq 2m - 1.$$

Together with Claim 44, this shows that the number of non-empty subsets of S of the form $B \cap S$, with $B \in \mathcal{B}_p$, is at most $2m - 1$.

It follows that $\Pi_L(S)$ contains the L non-empty parts R_i , $1 \leq i \leq L \leq q$, that are blocks picked by the p -adic balls, together with R_0 when R_0 is non-empty. A partition therefore corresponds to a combination of L blocks. We have $L \leq q \leq n$ and the number of blocks is at most $2m - 1$, hence there are no more than $\sum_{L=0}^n \binom{2m-1}{L}$ possible combinations of at most n blocks. The case $L = 0$ represents a partition $\Pi_0(S)$ with a single part $R_0 = S$ which is not picked by any of the q balls.

Next, we count the number of functions that can be constructed on a partition of S . Fix a partition $\Pi_L(S)$. On each non-empty part R_i of $\Pi_L(S)$, we define the values for $\hat{f}$ as follows: if R_0 is not empty then according to Proposition 39, for $x \in R_0$, we let $\hat{f}(x) := -\min_{1 \leq i \leq q} \hat{d}_{j_i}(x)$ where $j := [j_1, j_2, \dots, j_q] \in K^q$, and K is defined in (56), where $1 \leq q \leq n$. For a fixed $q \leq n$, there are M_ϵ^q possible choices of $j \in K^q$. Each such choice can be extended to a vector in K^n by repeating one of its components, without changing the minimum. Hence, as q varies over $1 \leq q \leq n$, there are no more than M_ϵ^n possible functions on R_0 .

If $x \in R_i$, $1 \leq i \leq L$, by Proposition 39, $\hat{f}(x) = \hat{d}_{j_i^\pi}(x)$, so for each R_i a single index from K is chosen. Hence there are no more than M_ϵ^L possible combinations of indices assigned to these L parts.

Thus, for a fixed value of L and a fixed partition $\Pi_L(S)$, there are no more than M_ϵ^{n+L} possible functions $\hat{f}$ on S . Since, for each L , there are at most $\binom{2m-1}{L}$ possible partitions corresponding to L blocks, the total number of functions is no more than

$$M_\epsilon^n \sum_{L=0}^n \binom{2m-1}{L} M_\epsilon^L.$$

C Proof of Theorem 43

We consider the bad event that (64) is *not* less than or equal to the sum of (65) plus some small deviation ϵ , that is, the event that

$$P(Yf(X) \leq 0) > P_m(Yf(X) \leq \gamma) + \epsilon.$$

This can be rephrased as the following event,

$$P(Yf(X) > 0) < P_m(Yf(X) > \gamma) - \epsilon. \quad (75)$$

Now, consider the bad event that there exists some $f \in \mathcal{F}_n$ and some value of $\gamma \in (0, 1]$ such that (75) occurs with $\epsilon := \epsilon(\gamma/2, \gamma\delta/2)$, where $\epsilon(a, b)$ is decreasing with a and b and is specified further below. Define the set $E(\gamma, \delta) \subset \mathcal{Z}^m$,

$$E(\gamma, \delta) := \{\xi : \exists f \in \mathcal{F}_n, P(Yf(X) > 0) < P_m(Yf(X) > \gamma) - \epsilon(\gamma/2, \gamma\delta/2)\}. \quad (76)$$

The probability of the bad event therefore equals $P^m\left(\bigcup_{\gamma \in (0, 1]} E(\gamma, \delta)\right)$.

We follow a technique of [37] (see [11, Lemma 15.5]). Let $\gamma_0 := \text{diam}(\mathbb{Z}_p) = 1$ and for $l = 1, 2, \dots, \infty$, define

$$\gamma_l := \left(\frac{1}{2}\right)^l$$

and

$$\Gamma_l := (\gamma_l, \gamma_{l-1}].$$

We have

$$\begin{aligned} P^m \left(\bigcup_{\gamma \in (0,1]} E(\gamma, \delta) \right) &= P^m \left(\bigcup_{l=1}^{\infty} \bigcup_{\gamma \in \Gamma_l} E(\gamma, \delta) \right) \\ &\leq \sum_{l=1}^{\infty} P^m \left(\bigcup_{\gamma \in \Gamma_l} E(\gamma, \delta) \right). \end{aligned} \quad (77)$$

Define the set

$$E_l := \left\{ \xi : \exists f \in \mathcal{F}_n, P \left(Yf(X) > 0 \right) < P_m \left(Yf(X) > \gamma_l \right) - \epsilon(\gamma_l, \gamma_l \delta) \right\}.$$

We claim the following.

Claim 45. For all $\gamma \in \Gamma_l$, $E(\gamma, \delta) \subseteq E_l$.

Proof of the Claim: For $\gamma \in \Gamma_l$ the statement $Yf(X) > \gamma$ implies $Yf(X) > \gamma_l$ so $P_m(Yf(X) > \gamma) \leq P_m(Yf(X) > \gamma_l)$. That $\epsilon(a, b)$ decreases with a implies $\epsilon(\gamma/2, b) \geq \epsilon(\gamma_l, b)$ for $\gamma \in \Gamma_l$. And because $\epsilon(a, b)$ decreases with b then $\epsilon(\gamma_l, (\gamma/2)\delta) \geq \epsilon(\gamma_l, \gamma_l \delta)$. It follows that $E(\gamma, \delta) \subseteq E_l$ for $\gamma \in \Gamma_l$. $\square$

Let

$$\epsilon(\gamma, \delta) := \sqrt{\frac{8}{m} \left(n \ln \left(\frac{16e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{2}{\delta} \right) \right)} \quad (78)$$

then

$$\begin{aligned} P^m(E_l) &= P^m \left(\exists f, P(Yf(X) \leq 0) > P_m(Yf(X) \leq \gamma_l) + \right. \\ &\quad \left. \sqrt{\frac{8}{m} \left(n \ln \left(\frac{16e^3 m^2 (4m-1)}{\gamma_l^2 n} \right) + \ln \left(\frac{2}{\gamma_l \delta} \right) \right)} \right) \\ &\leq \gamma_l \delta \end{aligned}$$

where the above inequality is obtained from Lemma 42. With Claim (45), it follows that the probability of the bad event (77) is bounded from above as follows

$$\begin{aligned} \sum_{l=1}^{\infty} P^m \left(\bigcup_{\gamma \in \Gamma_l} E(\gamma, \delta) \right) &\leq \sum_{l=1}^{\infty} P^m(E_l) \\ &\leq \delta \sum_{l=1}^{\infty} \left(\frac{1}{2} \right)^l \\ &\leq \delta. \end{aligned} \quad (79)$$

From (78) we have

$$\epsilon(\gamma/2, \gamma\delta/2) = \sqrt{\frac{8}{m} \left(n \ln \left(\frac{64e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{4}{\gamma\delta} \right) \right)}$$

and substituting this for $\epsilon(\gamma/2, \gamma\delta/2)$ in (76) means that the complement of the bad event is the event that (70) holds uniformly over all $f \in \mathcal{F}_n$ and all $\gamma \in (0, 1]$, which from (79) has probability at least $1 - \delta$.

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